Cell Membranes - Study Notes

1. The fluid mosaic model

The cell membrane is a phospholipid bilayer. Each phospholipid has a hydrophilic (water-loving) head and two hydrophobic (water-fearing) tails. In water, the tails hide inside while the heads face outward. Proteins float in this bilayer like icebergs, which is why it is called a mosaic.

2. Membrane proteins

Integral proteins span the whole bilayer and act as channels or carriers. Peripheral proteins sit on the surface and work as enzymes or signals. Carbohydrate chains attached to proteins form the glycocalyx, used for cell recognition.

3. Selective permeability

Small nonpolar molecules (O2, CO2) diffuse freely across. Ions and large polar molecules need transport proteins. Osmosis is the diffusion of water across a selectively permeable membrane, from low to high solute concentration.